COGNITIVE PHONETICS: THE UNIVERSAL PHONOLOGY-PHONETICS INTERFACE

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Abstract

Building on the theory of Substance-free Logical Phonology, which adopts the competence–performance dichotomy, strict internalism and modularity, we argue that phonology (phonological competence) and phonetics (the sensorimotor system in charge of speech production and perception) are distinct and non-overlapping domains in the complex process of the externalization of language through speech. These two domains are connected via the phonology-phonetics interface (PPI): a bidirectional transduction system that converts between feature-based surface phonological representations and the temporally coordinated motor programs that we call phonetic representations. This transduction is captured by a theory of the PPI called Cognitive Phonetics, whose workings are presented in this chapter. We apply this theory in the analysis of some notorious sound patterns, arguing that there is no need to assume the existence of language-specific phonetics.

Keywords: *phonology-phonetics interface*, *Substance-free Logical Phonology*, *Cognitive Phonetics*, *competence vs. performance*, *modularity*

1. Introduction

Substance-free Phonology (SFP) is a research program for the scientific study of phonological knowledge. Its fundamental assumption is that the content that comprises phonological knowledge – phonological representations and computations – is devoid of phonetic substance. **Phonetic substance** corresponds to the articulatory, acoustic and auditory properties and processes of speech. So, speech-related information plays no *direct* role in the internal workings of phonology, and the totality of phonology is contained within the human mind. Although the modern generative conception of SFP emerged with Hale & Reiss (2000a; 2000b), it has since proliferated into many different, more or less incompatible theories, all of which share the above stated fundamental assumption (see Chabot, to appear, and papers in Chabot, 2022). One of these theories is called **Substance-free Logical Phonology** (SFLP) (Bale & Reiss 2018; Volenec & Reiss 2020; Reiss 2021). As the name implies, its tenet is that substance-free phonological knowledge can productively be modeled using set-theoretic logical operations such as set subtraction, unification, and intersection.

By excluding phonetic substance from phonology, and thus maintaining that phonology and phonetics are distinct non-overlapping domains, SFLP and other substance-free theories invite the following questions: What exactly is the relationship between phonology and phonetics? What bridges the gap between the purely mind-internal phonological knowledge and the peripheral physiology of speech? How is phonological knowledge deployed in actual acts of speech production and perception? These are the questions that define the study of the **phonology-phonetics interface** (PPI).

In this chapter, we provide tentative answers to these questions from the perspective of SFLP. We begin by laying out some basic assumptions about the nature of phonological knowledge and its place in the human mind, thus allowing for a more precise definition of the PPI (§2). Then we present a particular theory of the PPI called Cognitive Phonetics (§3). Finally, we apply this theory in the analysis of some notorious sound patterns, arguing that there is no need to assume the existence of language-specific phonetics (§4).

2. The phonology-phonetics interface

We take the phonology-phonetics interface (PPI) to be the connection between **phonological competence** (or **phonology** for short) and the **sensorimotor** (SM) **system**[1] (or **phonetics** metonymically). Naturally, the characterization of this interface will greatly depend on the theoretical assumptions that we make about the two involved systems, especially about the phonological side of the interface since the working of the SM system are much better understood. In (1), we therefore briefly state what our starting assumptions are, and then in the remainder of this section we provide a more precise definition of the PPI building on those assumptions.

(1) Some relevant principles of Substance-free Logical Phonology

- **Competence vs. performance**. We strictly differentiate between linguistic knowledge (competence) and the use of that knowledge in communication (performance). Performance always utilizes competence, but it is determined by many additional factors that are unrelated to competence, such as memory limitation, attention, mood, recent experiences, individual anatomical properties, etc. Since performance is a confluence of many different factors, it never reflects competence directly, and should not be equated with it. Phonological knowledge – the subject matter of phonology – is part of linguistic *competence*. Speech – the subject matter of phonetics – is one kind of linguistic *performance*.
- **Internalism**. The entirety of linguistic competence is contained within the mind/brain. Phonological units, operations and patterns exist only in the human mind/brain and not 'out there' in the (extra-mental) world. Phonology (as a science) is therefore not about speech or verbal behavior; it is about linguistic competence – a crucial distinction. Speech serves as one *source of evidence* for phonological competence because every act of speaking necessarily utilizes this mind-internal knowledge.
- **Formalism**. All aspects of phonology are substance-free. The term 'substance' refers to the spatial and temporal aspects of the production, perception and acoustics of speech. Phonological

[1] Since the term *sensorimotor system* is used ambiguously in the linguistic literature, it should be noted that we use that term to denote a large-scale brain network that includes the pre-central gyrus, the post-central gyrus, and the supplementary motor area.

competence is to be characterized as formal – that is, explicit, logically precise, and substance-free – manipulation of symbols.

- **Realism.** To the extent that they are correct, the postulates of SFLP such as features and rules are actually instantiated in the brain. They are characterized by so-called 'psychological reality'.[2] Dualism – the idea that mind and matter are categorically different and mutually incommensurable – is rejected. The mind is an abstract characterization of some subset of brain functions. We sometimes speak of mental functions *without* reference to the brain merely because of our limited understanding of the brain, not because we subscribe to any form of dualism.

- **Nativism**. The atomic, indivisible elements of phonological competence are innate. The human body, which necessarily includes the brain where linguistic competence is stored, is a product of the interaction between the genotype, the environment, and epigenetic factors. Structures that are determined by the genotype and not exclusively by the other two factors, are said to be innate. Since the interplay of nature *and* nurture determines the structure of the entire body, there is no reason *not* to assume that the same principle holds for the structure of those parts of the brain that are responsible for cognition. Thus, some aspects of cognition are innate and some are learned. Learning consists of making new combinations from pre-existing innate elements. The most primitive atomic elements and operations cannot themselves be learned because they are precisely what is needed for learning to take place. The most primitive atomic elements of phonological cognition, on the basis of which mature phonological competence is built, are features.[3] It follows, then, that features are innate, and that learning in the domain of phonology consists of building complex representations on the basis of those innate features and perhaps on the basis of the innate format (but not the exact content) of rules.

Especially important among these assumptions is the competence–performance distinction: we strictly maintain that phonology is a part of linguistic competence, while phonetics is a type of linguistic performance. Our view of the relationship between phonology and phonetics is built on this premise, and this view can concisely be stated in the following way. Phonological competence consists of discrete and timeless symbols and formal operations. On the other hand, speech is characterized by gradient, temporally arranged movements of speech-related organs and the concomitant continuously varying sound waves. Since humans do speak, it is a logical necessity that phonological competence successfully relays information to the SM system, which is in charge of speech production. The brain-internal connection between these two systems constitutes the **phonology–phonetics interface** (**PPI**). Clearly, the notion of an interface presupposes the existence of two *separate* entities (Figure 1).

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[2] The term is taken from Chomsky (1980: 107): "The question is: what is 'psychological reality,' as distinct from 'truth, in a certain domain'? I am not convinced that there is any such distinction, and see no reason not to take our theories tentatively to be true at the level of description at which we are working, then proceeding to refine and evaluate them and to relate them to other levels of description, hoping ultimately to find neural and biochemical systems with the properties expressed in these theories."

[3] The atomic elements of phonological cognition possibly also include tonal units, components of syllable structure, etc. We focus here solely on features mostly for the sake of simplicity and expository ease, but our point applies to *all elementary* aspects of phonology, whatever they exactly turn out to be.

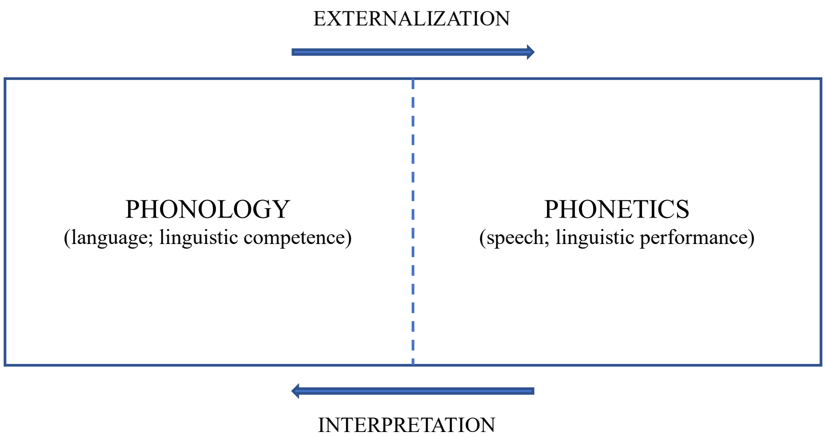

Figure 1. The phonology-phonetics interface (PPI) in the theory of Substance-free Logical Phonology (SFLP).

The dashed line between phonology and phonetics in Figure 1 represents the *boundary* between them. The idea, then, is that phonology is about (one aspect of) *language* or *linguistic competence*, and phonetics is about one kind of *linguistic performance*, namely *speech*. There is no overlap between them, and the boundary is strict and fixed. Because no part of phonetics is a part of language, there is no such a thing as 'language-specific phonetics', as we further elaborate in section 4.

Clearly, our approach to the PPI also assumes that the mind is modular. A **module** is a specialized cognitive system governed by principles not completely shared by any other part of cognition. Each module consists of a finite set of primitive, atomic symbols (basic **representations**) and a finite set of operations that manipulate those symbols (basic **computations**). In other words, a module performs a specific set of computations over a specific set of representations. Furthermore, modules can have their own encapsulated and interacting components, which we may call **submodules** – modules within modules (cf. Curtiss 2013). Language (linguistic competence, I-language) is a module of the human mind (Chomsky 1984; Smith 2011; Allott & Smith 2021). Phonology (phonological competence) is a submodule of language. Its basic representations are phonological features (see Reiss & Volenec 2022), and its basic computations are set-theoretic rules (see Reiss 2021).

Despite the fact that different modules have different computational and representational parts, all modules have certain universal characteristics (Fodor 1983; Boeckx 2010: 125–127; Smith & Allott 2016: 17–30). These are summarized in (2).

(2) Universal characteristics of cognitive modules

- **Domain specific**. Every module consists of a unique combination of computations and representations. Each has a specific domain of operation, such vision, audition, language, etc.

- **Informationally encapsulated**. A module does not have direct access to another module. Blind to the workings of other modules, it operates independently. Modules communicate indirectly, through interfaces.
- **Fast and automated**. Since they are encapsulated and therefore need only to consult a restricted range of options, they operate relatively quickly. They compute in an automatic and obligatory fashion, so they cannot be influenced by conscious command.
- **Innate**. The general structure of a module is innate, determined biologically, and thus uniform across the species. This is why modules have a fixed developmental schedule. The particular peripheral aspects of modules are determined by experience during the course of the maturation of the organism.
- **Hardwired**. A cognitive module corresponds to a dedicated neural network,[4] and is therefore subject to selective pathological breakdown, manifested as double dissociations.[5]

Although modularity tends to be discussed at a fairly abstract mentalistic level, it is possible to discuss modularity at the level of the brain, at least to a certain degree, due to discoveries in contemporary neurobiology. In this context, modules are independent, specialized neural circuits that perform unique functions. The brain's modules consist of neurons that are more connected to one another than to neurons that comprise other modules (Bassett & Gazzaniga 2011). In *Rules and Representations* (1980), Chomsky controversially designated the language faculty and other modules as "mental organs" (p. 39), claiming that if it were not for these highly specialized mental organs, whose general structure is due to innate endowment, "individuals would grow into mental amoeboids, unlike one another, each merely reflecting the limited and impoverished environment in which he or she develops" (p. 46). In the subsequent decades, research in cognitive neuroscience and clinical neurology has vindicated this claim. Gallistel (1996: 86), for example, points out that:

> "Chomsky was right to argue that there is a module for learning language. In fact, I think he was right to call it an organ, because the analogy to the specialization of structure and function that we see in organs is an appropriate way to think about the specialization of structure and function we see in the various learning organs. The language organ is just one example among many. These organs within the brain are neural circuits whose structure enables them to perform one particular kind of computation. This view is the norm these days in neuroscience."

Crucially for the discussion at hand, different modules of the mind/brain communicate via **interfaces**: configurations in which the output of one module serves as the input to another module. Manipulation (e.g., reordering, regrouping, deletion, addition) of representational elements *within* a module, and *without* a change in the representational vocabulary is called **computation**.

[4] We are using the term 'neural network' in the literal, neurobiological sense: a network of actual neural cells in the human brain. We are not referring to artificial neural networks from the field of machine learning.

[5] "Double dissociations are, in fact, the holy grail in neuropsychological research, since they often provide compelling evidence that two tasks rely on at least partly segregated cognitive mechanisms" (Kemmerer 2015: 31). Curtiss (2013) provides a host of examples from experimental research where it was demonstrated that language can be doubly dissociated from other cognitive systems, and also provides some evidence for the internal submodularity of language, particularly dissociations between phonology and phonetics.

Conversion of an element in one representational vocabulary into a distinct vocabulary, i.e., a mapping between dissimilar *types* of representations, is called **transduction** (Pylyshyn 1984). So, modules compute, interfaces transduce. For example, phonology is a module that computes over a single representational vocabulary, the set of phonological features, without the possibility of changing features into, say, sound waves. And due to informational encapsulation, phonological computation proceeds independently of what goes on in syntax (a different (sub)module), or in the visual system (another module), or in the mouth (not a module), or anywhere else. On the other hand, our proposed theory of the PPI, which we survey in section 3, is an example of a transducer. Specifically, it is a transducer that connects phonology and the sensorimotor system, functioning as an interface between them. It converts features, which are native and recognizable only to phonology, into a representational format (called a phonetic representation) that is native and recognizable to the SM system (but no longer to phonology). As we can see, an interface between two organic systems such as cognitive modules is defined by the form of the input, the form of the output, and a set of transformations that relate them. At the level of the brain, computation and transduction are distinguished by differences in the parameters of neural activity, such as the localization of action potentials (place code), their relative timing (temporal code) and the frequency of their repetition (rate code).

The representational part of the phonological module is comprised of features. In line with the principles in (1), features are mind-internal units and as such they are necessarily substance-free, that is, they *do not* contain information on the temporal coordination of muscle contractions, on the spectral configuration of the acoustic target to be reached, and so on. Yet without this information, the respiratory, phonatory and articulatory systems cannot produce speech in real time (Lenneberg 1967; Levelt 1989). The motor system for speech production requires information about substance and time in order to arrange the articulatory score (see below for elaboration), therefore this information has to be integrated into a representation before being fed to the motor system. A simple and plausible way to bridge this gap is to posit a cognitive phonetic transduction system that converts feature matrices into true phonetic representations that provide the SM system with legible information needed to produce speech. This is indeed an expected result of modularity: since phonology is informationally encapsulated, it can only communicate with the SM system through a bidirectional interface.

We can summarize the discussion so far thus:

- Outputs of the phonological module, surface phonological representations (SRs) consisting of substance-free features, do not contain articulatory, auditory and temporal information.
- The SM system requires articulatory, auditory and temporal information in order to produce speech.

∴ Phonology does not communicate with the SM system directly; SRs are not legible to the SM system due to modularity.[6]

∴ The interface between phonology and the SM system is mediated by transduction.

The nature of the transduction that occurs at the PPI is explained by the theory of Cognitive Phonetics, to which we now turn.

3. Cognitive Phonetics

Cognitive Phonetics (CP) is a theory of the phonology-phonetics interface built on the tenets of Substance-free Logical Phonology (Volenec & Reiss 2017; 2020; Reiss & Volenec 2022). CP proposes that the PPI consists of at least two **transduction procedures** that convert the substance-free output of phonology into a representational format that contains substantive information required by the SM system to externalize language through speech.

The inputs to CP are the outputs of phonology, i.e., surface phonological representations (SRs). SRs are strings of segments, each of which is a set of features. Each feature of SRs is transduced and subsequently receives interpretation by the SM system. This transduction is carried out by two algorithms (cf. Marr 1982: 23–24). The **paradigmatic transduction algorithm** (**PTA**) takes a feature (a symbol in the brain) and relates it to a motor program which specifies the muscles that need to be contracted in order to produce an appropriate acoustic effect. The **syntagmatic transduction algorithm** (**STA**) determines the temporal organization of the neuromuscular activity specified by the PTA. In simpler terms, PTA assigns muscle activity to each feature, STA distributes that activity temporally. These transduction algorithms yield an output representation of CP, which then feeds the SM system. The output of CP is called the **phonetic representation** (**PR**),[7] and it can be defined as a complex array of temporally coordinated neuromuscular commands that activate muscles involved in speech production.

The traditional schema of phonological competence can now be expanded to accommodate the transduction performed by CP, transforming it into a more complete 'speech chain' in Figure 2. The gray parts of the schema represent phonological competence, while the black parts correspond to the initial phonetic steps in speech production. That is, the difference in shading parallels the competence–performance dichotomy: phonology is competence, cognitive phonetics is (one component of) performance. Figure 2 shows that what is loosely referred to in the literature

[6] Here we are more concerned with the consequences of modularity on the nature of the PPI, but it is worth pointing out that the same line of modular reasoning partly explains why phonology has to be substance-free: informational encapsulation makes it impossible for phonetic substance to influence phonological computation. The SM system can't see the inner workings of phonology, so transduction at the interface is needed; likewise, phonology can't see the workings of the SM system, so there can be no phonetic substance in phonology.

[7] In generative linguistics literature, the output of phonological computation, what we call a 'surface phonological representation', is sometimes called a 'phonetic representation'. Note that we use the term very differently—our phonetic representations do not contain the phonological features that are in URs and SRs, and they do not belong to language/grammar.

as ‘the phonetic form’ (PF) or ‘the externalization of language’ (EXT) (Chomsky 1986: 68; Chomsky et al. 2017: 16) has a sophisticated internal structure.

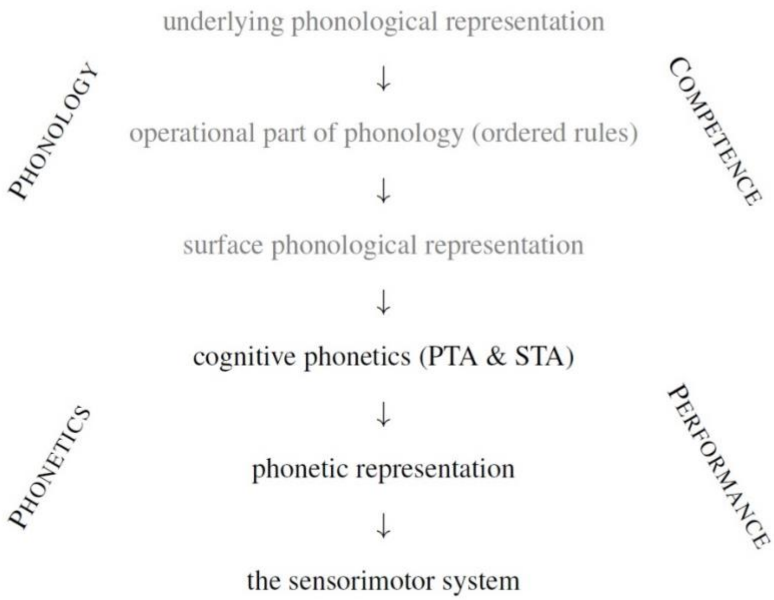

Figure 2. Phonology and phonetics are mediated by cognitive phonetics.

We can start clarifying the effects of the PTA and the STA by first exploring the fate of an isolated transduced feature. Take, for example, the feature +ROUND. We assume that the phonological module outputs, as part of its surface representation, the feature +ROUND, which then serves as an input to CP’s transduction procedures. The PTA takes +ROUND and assigns neuromuscular activity to it, specifying which muscles need to be contracted for its appropriate articulation. In this particular case, where the intended effect is lip rounding, at least four muscles will be engaged: *orbicularis oris*, *buccinator*, *mentalis*, *levator labii superioris* (Seikel et al. 2020: 825–826). The STA then specifies the temporal aspects of the PTA’s effects. For every muscle contraction determined by the PTA, the STA determines for how long these contractions should be maintained, and in which relative order. Thus, the PTA determines which muscles need to be contracted to realize lip rounding, and the STA determines when and for how long these contractions need to be maintained. The joint result of the PTA and the STA applying to a phonological feature is a **transduced feature**, which we take to be **the basic unit of speech production**. For notational purposes, we can refer to a transduced feature as ‘$\mathbf{PR_F}$’, where ‘PR’ stands for ‘phonetic representation’ (the output of CP), and ‘F’ stands for any feature. We can now say that the general function of CP’s transduction operations is to perform the following conversion: $\mathbf{F \rightarrow PR_F}$. As already noted, F’s are neural symbols that participate in phonological computation. PR_F’s are the output of CP that serve as basic units of speech production, as they can be executed by the primary motor cortex. We can therefore understand each PR_F as an array of temporally coordinated neuromuscular commands that activate muscles involved in speech production. Note that the F $\rightarrow$ PR_F conversion is transduction and not computation since this mapping is between entities that belong to different vocabularies: PR_F's are unrecognizable to phonology and they are not what phonology computes over; also, F's are unrecognizable to the primary motor cortex and they are not what the SM system realizes in speech. Phonology works with features, while the SM system realizes *transduced* features.

Let us now introduce a slight complication in our exploration of the transduction of +ROUND. Since lip rounding is known to have systematically varying muscular expression due to interactions with other features form the same segment, the transduction algorithms have to be able to account for this systematic variation. In the pronunciation of a high back vowel [u], +ROUND will be realized with a protruding motion of the lips, whereas in a high front vowel [y], +ROUND will be realized with a compressing motion of the lips (Catford 1982: 172–173). Thus, we must assume that PR_{+ROUND} is different for [u] than for [y]. Specifically, in the former case the PTA and the STA will assign simultaneous contraction of the superior and inferior parts of *orbicularis oris*, contraction of *mentalis* (for protruding the lower lip) and *levator labii superioris* (for protruding the upper lip), and relaxation of *buccinator*. In the latter case, PR_{+ROUND} in addition to contracting the aforementioned muscles will also include a compressing movement (lips drawn together horizontally) caused by the contraction of *buccinator*. The difference between protrusion and compression in PR_{+ROUND} is dependent on whether PR_{+ROUND} is interacting with PR_{+BACK} or $PR_{–BACK}$. So we can see that the effect of the PTA will necessarily take into account the specification of other features in the segment that it is transducing. In fact, therein lies the reason why this algorithm is termed *paradigmatic*: it considers *all of the features within a single segment* when determining the required neuromuscular schema for every feature.

Let us now assume that CP is transducing +ROUND from [u], but [u] is preceded by [s], like in the word *soup*. What we invariably find in cases like this is that the preceding consonant is rounded even though its own specification for ROUND is negative, i.e., [s] contains –ROUND (Volenec 2023). In this case, when determining the temporal parameters for the PR_{+ROUND} originating from [u], the STA will extend PR_{+ROUND} in the anticipatory (regressive) direction to influence $PR_{–ROUND}$ from [s]. This amounts to the familiar notion of anticipatory coarticulation. Indeed, **coarticulation** is the principal phenomenon that arises at the PPI, and any successful PPI theory must explicitly account for it. With its two transduction algorithms, Cognitive Phonetics opens the possibility of elegantly accounting for subtle yet systematic interactions of various kinds of coarticulatory effects. We can now also see why this other algorithm is called *syntagmatic*: it considers the features from a particularly ordered *sequence of segments* when determining the required timing of neuromuscular schemata. This is in accordance with the need for a "central plan model" of speech planning, as discussed by Lenneberg (1967: 106).

CP's transducers are analogous to other transducers in the body, such as those involved in audition or vision, in the sense that they are part of the human biological constitution. Their maturation is determined and guided genetically, and merely triggered by experience. They do not entail learning nor do they imply knowledge (e.g., 'knowing how' to speak). They are also not part of the language faculty, and therefore are not part of Universal Grammar (UG), even though they are **innate**. It is for this reason that we do not associate CP with what has traditionally been called 'Universal Phonetics' (e.g., in Chomsky & Halle (1968) and related work) – we wish to avoid the incorrect interpretation that CP is part of UG. It is not, just as the interface between the language faculty and the visual system, which is utilized in reading and writing, is not part of UG.

Like any other kind of transduction in the body, such as the various signal transductions that take place in the ear, CP's transduction is also **automatic** and **deterministic**. This means that CP assigns the same neuromuscular schema to each feature every time that feature is transduced. However, this *does not* mean that CP's outputs, phonetic representations, should be equated with actual articulatory movements or with the acoustic output of the human body. What is actually pronounced is further complicated in the process of language externalization by a great number of factors. Transduction is accompanied by other performance factors that have no bearing on either phonology or transduction, factors like muscle fatigue, degree of enunciation, interruptions due to sneezing, trying to achieve a certain intensity level, being in a certain emotional state (angry, sad, afraid, relaxed, excited, etc.), and many other situational and pragmatic effects, all of which will have an influence on the final output of the body, and will therefore make (co)articulatory variation seem even greater. For that reason, it is not the case that the articulatory and the concomitant acoustic substance will always be identical for each feature or feature combination. But this apparent **lack of invariance** in the realization of a cognitively invariant category is not a matter of transduction, but rather is a result of intractable performance factors. For example, if we are externalizing the substance-free feature +BACK, then we expect that during the pronunciation of a vowel that contains that feature the tongue dorsum will typically be positioned in a certain posterior space in the oral cavity and we also expect its F2 to be in a certain low region. Every repetition of what is mentally that same vowel will lead to correlates that fall within these spaces (barring particularly strong coarticulatory effects), even though precise measurements will show that no two repetitions are physically identical. The automatic, deterministic nature of transduction is what ensures that the vast majority of feature correlates fall within relatively predictable phonetic spaces, while the tremendous complexity of all other performance components gives rise to lack of invariance.

It is also clear that the outputs of CP are subject to further manipulation by mental systems that are in charge of conscious action, creativity, free will, intentionality, and other relatively poorly understood abilities. As self-aware and intelligent beings, we humans can do whatever we want once CP's representations reach the level of consciousness. We can use these phonetic representations to shout in anger, whisper seductively, argue with increased diction, sing in falsetto, imitate Darth Vader's voice, and so on. All of these deliberate acts will modify articulation and change the final acoustic output from the body, but they have nothing to do with phonology or Cognitive Phonetics. To a certain extent, these acts are governed by activity in the prefrontal cortex (for executive functions like planning and decision-making) and in the primary motor area (for voluntary motor control), but their general nature is still quite mysterious (Poeppel et al. 2021).

Since they are part of the performance machinery and not of competence, the transduction algorithms are not specific to any particular I-language. Thus, CP will transduce SRs in a uniform manner irrespective of whether these SRs come from an I-language of an 'English' speaker, or a 'Croatian' speaker, or a '!Xhosa' speaker. Notice, however, that while the transduction system itself is invariant, the system's outputs (PRs) are going to be different from one another insomuch as they reflect different language-specific phonological SRs. So we can say that PRs are surface-representation-dependent since CP will automatically transduce any SR that is fed to it and any particular I-language will generate its own distinct set of SRs via its language-specific phonology. For example, English SRs will not have click segments in them, but some !Xhosa SRs will have clicks. But for CP itself it is irrelevant whether an SR comes from an English or a !Xhosa speaker – it will receive the same treatment by the PTA and the STA. This is analogous to how the organ of Corti in the inner ear would treat all spoken signals uniformly irrespective of whether they are 'English' or '!Xhosa'. Clearly, then, the combination of Substance-free Logical Phonology and Cognitive Phonetics leaves no room for 'language-specific phonetics' (also consult Figure 2), which brings us to the next topic.

4. Language-specific phonetics does not exist

Our two theories of the externalization of language preclude **language-specific phonetics** (**LSP**). We argue that anything that has to do with the *mental representation* (knowledge) of the sound patterns of *language* is phonology. We also think that alleged language-specific phonetic patterns can be reanalyzed so as to conform to our model, which only contains language-specific substance-free phonology and non-language-specific cognitive phonetics. Let's take a closer look at these claims.

Our rejection of LSP is first and foremost a result of consistently adhering to the competence–performance dichotomy (see (1)). Something can be language-specific only by virtue of belonging to an individual's linguistic competence, i.e., to their I-language. Something that is not part of an I-language cannot be specific to an I-language, i.e., it cannot be language-specific. Since speech, the subject matter of phonetics, is not language (competence, I-language), no part of it can be language-specific. The idea of 'language-specific phonetics' is thus a contradiction in terms just like, say, the idea of 'language-specific vision', because neither vision nor speech are part of linguistic competence. Both the visual system and the articulatory-auditory system *interface* and *communicate* with linguistic competence—vision in reading, the articulatory-auditory system in speaking—but neither is *part of* competence. Thus, if we are doing generative linguistics where the object of study is linguistic competence, it is theoretically more coherent not to postulate LSP.

Another reason to reject LSP is scientific simplicity. It has repeatedly been argued in the literature that particular phonological analyses become much simpler if we assume the existence of LSP, which then takes some of the burden off of phonology. If, on the other hand, we assume that LSP does not exist and that phonology must account for the entire phenomenon in question, the analysis of the same data-set becomes less elegant. But this kind of reasoning is problematic. Indeed, a *particular* phonological analysis can become simpler if we shift a part of it into an LSP module, but now the *whole general theory* of the externalization of language is much more complicated because we are assuming a separate cognitive module with its own architecture, in addition to other necessary components. Frameworks that admit LSP consist at least of the following components: phonology; phonology-phonetics interface; language-specific phonetics; interface between language-specific phonetics and the sensorimotor system; the sensorimotor system. Our proposed framework, on the other hand, assumes only these components: Logical Phonology, Cognitive Phonetics and the SM system. Clearly, more is gained in terms of scientific parsimony if the simplicity principle is applied to the entire explanatory model rather than to individual analyses: it is better to keep the overall model of the externalization of language simple than to simplify the analyses of some specific phonological data-sets while dumping the rest into LSP.

Of course, simplicity itself does not tell us if a theory is true or not. So, the main question is: can the simpler model that we are proposing account for the relevant empirical data, i.e., for the sound patterns that have motivated the positing of LSP? For ease of exposition, we can split the answer into two parts. First, we address the arguments that have been brought forward as justification for postulating LSP. Second, we show how our simple SFLP + CP framework deals with these alleged LSP patterns.

The reason why LSP was postulated in the first place is concisely summarized in a recent encyclopedic article on phonetics:

> "Early theorists assumed that phonetic implementation of phonological features was universal, but it has become clear that languages differ in their phonetic spaces for phonological elements, with systematic differences in acoustics and articulation. Such language-specific details place phonetics solidly in the domain of linguistics; any complete description of a language must include its specific phonetic realization patterns." (Whalen 2019)

The most prominent of these "early theorists" are Chomsky and Halle, who claimed that phonology and phonetics are strictly separated, and that no part of phonetics is language-specific (despite the fact that they used terms like 'phonetic representation' and 'phonetic features' within phonology; for them 'phonetic' simply meant 'surface phonological'). After *SPE*, an increasing number of data-sets were being interpreted as incompatible with the *SPE* model because neither phonological rules that manipulate discrete features nor universal phonetic interpretation could account for them. To solve this problem, another module was introduced into generative grammar: language-specific phonetics. The most influential early set of examples of LSP patterns, which included vowel duration and voicing timing, were presented and analyzed by Keating (1985).[8] Keating argued that

[8] See also Pierrehumbert (1980) for proposals about (what was later named) 'generative phonetics', as it applies to the phonetic implementation of English intonation.

vowel duration and voicing timing systematically differ from language to language; therefore, they cannot be a result of universal phonetic interpretation and they have to be "cognitively represented" and "under explicit control by the speaker" (p. 128). But since they are gradual and thus cannot be captured by existing discrete phonological features, they cannot be part of the phonology. Keating's solution was to add another rule-based component to the grammar – the phonetic module – which accounts for such fine-grained sound patterns.[9] Subsequent studies have greatly expanded the inventory of such putative LSP patterns: acoustic spaces of vowels (Bradlow 1995), vowel duration (Laeufer 1992), vocal fold vibration (Kingston & Diehl 1994), voice onset time (Cho & Ladefoged 1999), palatalization patterns (Zsiga 1995), positioning of the articulators before initiating speech (Gick et al. 2004), prosodic focus (Peters et al. 2014), and so on. The current consensus appears to be that there is "overwhelming evidence for 'language-specific phonetics'" (Whalen 2019), and that "the result of a century of phonetic work was to establish phonetics as a module of linguistics [i.e., grammar – cr & vv] equally important to the phonological module" (Zsiga 2020: 19).

However, in what follows we argue that this mass of alleged LSP patterns arose because the research methods used to gather and analyze the data were not aligned with the fundamental principles of generative linguistics, and thus the data were not properly interpreted.

Pierrehumbert et al. (2000: 286) refer to Bradlow (1995) as an example of a study that shows that it is necessary to assume the existence of LSP. Bradlow (1995) compared the acoustic realization of vowels in 'General American English' and in 'Madrid Spanish', trying to determine if the realization of the vowels from these 'two languages' is governed by universal or language-specific principles. Four speakers of English and four speakers of Spanish participated in the experiment, each reading a list of carrier sentences five times, yielding 20 tokens for every word (like *beat*, *bit*, *bet*, etc.) that contains a particular vowel. F1 and F2 values were extracted and then averaged across each of the 'two languages'. Formant values for four vowels – [i, e, u, o] – were then plotted on an F1-by-F2 graph, which is shown in Figure 3.

[9] Notice that another way to solve this problem was to revise the specifics of the existing components of generative grammar and its interface with the SM system. This is ultimately the avenue that we are exploring with Cognitive Phonetics.

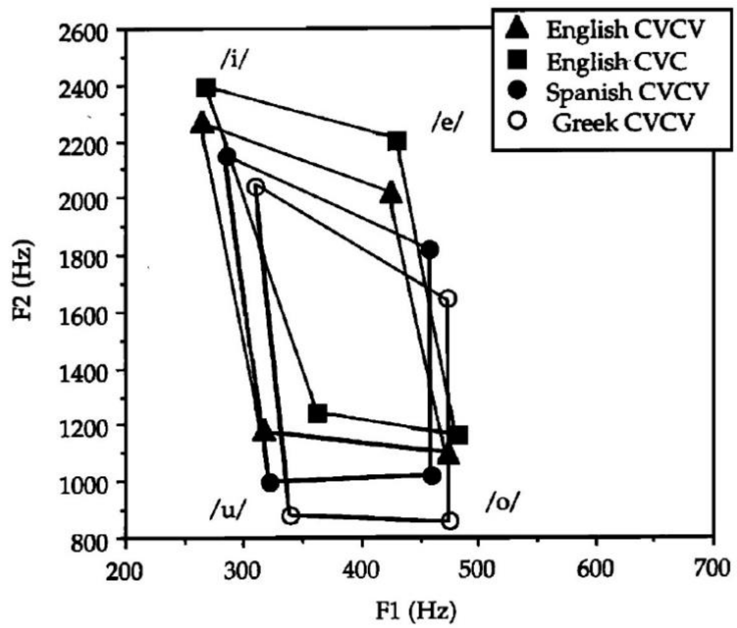

Figure 10. Comparison of the areas in the F1 x F2 space covered by English (in CVC and CVCV contexts), Spanish, and Greek [i], [e], [o] and [u]. The graph is reproduced from Bradlow (1995: 1919). Greek data is from Jongman et al. (1989).

Discussing the results that are graphically represented in Figure 3, Bradlow (1995: 1920) offers the following interpretation:

> "The crucial point in this comparison of the English and Spanish vowels is that all the vowels of English are generally higher in F2 than their Spanish counterparts. […] This comparison of vowel locations across English and Spanish indicates the presence of a language-specific factor in the phonetic realization of phonemic vowel categories."

Similarly, comparing Spanish and Greek (with Greek data originating from Jongman et al. 1989), Bradlow (1995: 1920) says that

> "these *two phonemically equivalent vowel systems* show a systematic difference regarding the acoustic realization of the shared vowel categories: the vowels of Spanish are generally higher in F2, and lower in F1 than the corresponding vowels of Greek. Thus we can conclude that a language-specific, base-of-articulation property plays and important role in determining the location of vowel categories in the acoustic space" [emphasis added]

It is important to emphasize that Bradlow (1995: 1922) makes an *a priori* assumption that the languages under study have identical vowel systems, not even entertaining the possibility that the differences in phonetic realizations of vowels are due to phonological (featural) differences:

> "This cross-linguistic difference has been accounted for by the notion of a language-specific base-of-articulation property, which is an important aspect of the description of the sound system of a language, and serves, in part, to differentiate systematically the general phonetic quality of two languages that many share certain phonemic categories. In other words, vowel categories that have the same phonological feature specifications and that occupy similar positions in the acoustic space

> across two different languages may have different precise phonetic realizations due to different bases-of-articulation of each language"

A fundamental problem in studies that argue for the existence of LSP, such as Bradlow (1995), is that they draw conclusions about the general architecture of I-language (recall that 'I' means *internal* to an *individual*) on the basis of data that come from *multiple people's linguistic performance.* Why is this a problem? First, even the speech production of *one person* does not reflect the properties of that person's I-language *directly*, since speaking entails the interaction of a vast number of cognitive and non-cognitive systems. Sorting out which system contributes to which aspect of the data and inferring the properties of a mental grammar from spoken utterances are notoriously difficult tasks, despite the fact that speech is by far the best source of evidence for that. And second, the conflated parameters of the speech production of multiple speakers (such as formant values averaged across a group of participants in an experiment) are even farther from reflecting the properties of any one I-language. Traditional everyday labels such as 'English' make it all too easy to believe that we are dealing with 'one language' when in fact we are neither dealing with one entity nor with 'language' in the generative-cognitive sense of that term. In the study of the architecture of the human language faculty, there is no 'English'. There are only individuals, each of which has a unique I-language in their brain. No two humans have identical I-languages, but they can communicate to the extent their I-languages are sufficiently similar. For practical, political, social, judicial, cultural and other purposes we can group somewhat similar I-languages together, calling them 'English', 'General American English' or whatever, but if we are concerned with the architecture of the human mind, we must not confuse these labels for our actual object of study.[10] If we conflate data from multiple different I-languages, we are inevitably obscuring important properties of each individual system.[11] Therefore, what Bradlow (1995) is dealing with are not 'two languages' – English and Spanish. They are actually eight different I-languages (since there were eight participants in the experiment), and so we should not *a priori* (on the basis of nothing more than intuition or convention) assume that the phonological components of these I-languages contain the same vowels. The *phonological* properties of these I-languages must be established by careful analysis *before* drawing conclusions about the "phonetic realization" of the "phonemic vowel categories" (Bradlow 1995: 1920). The phonetic differences in question are easily accounted for if we assume that the vowels are actually phonologically (featurally) different, which automatically removes the need for any kind of "language-specific base-of-articulation property", that is, LSP.

It is interesting that Bradlow's own data illustrates the danger of conflating data from multiple I-languages. In Figure 3, the acoustic spaces of 'English' vowels in CVC words differ as

[10] Within generative linguistics, it is innocuous enough to use terms like 'English' to give a quick and rough designation of the type of grammar that one is dealing with, so that other linguists can get an approximate sense of what kind of data and patterns to expect. But it is far more problematic to claim to be dealing with the universal architecture of mental grammar while actually describing the verbal behavior of multiple people.

[11] This has proven to be especially problematic for extremely high-level generalizations, such as the so-called 'Indian English language', a label that conflates hundreds of millions of I-languages. See Pisegna & Volenec (2021) for discussion.

much from ‘English’ vowels in CVCV sequences as they do from ‘Spanish’ vowels. So, within-‘language’ differences are just as great as between-‘language’ differences. Bradlow attributes the *between*-‘language’ differences to a separate module of grammar, language-specific phonetics. But if the between-‘language’ differences warrant a postulation of an entirely separate component of grammar, then why not attribute the *within*-‘language’ differences, which are just as great, to a further specialized component of grammar? Given Bradlow’s data, why don’t we postulate one module for the phonetic implementation of CVC words, another module for the phonetic implementation of CVCV words, etc.? We will present our explanation of these data shortly, but the point here is that this problem arises because of a failure to distinguish between competence and performance, because of an unwarranted conflation of different I-languages,[12] and because of unjustified *a priori* assumptions about the phonological inventories of these I-languages. In short, the methods that were used to gather and analyze the data were not aligned with the fundamental principles of generative linguistics. This kind of misalignment between method and theory can be found in all studies that argue for the existence of LSP – it is the very mistake that gave birth to LSP in the first place.

Our LP + CP model offers a simple explanation of the patterns under consideration: ‘English’-type vowels are *featurally* somewhat different from ‘Spanish’-type vowels, which is why in production they occupy different acoustic spaces; ‘English’-type vowels in CVCV words are pronounced differently from those same vowels in CVC words due to V-to-V coarticulation (which is captured by CP) which is present in the former but not in the latter case. There is no language-specific phonetics; its postulation plays no explanatory role; its invention is not warranted.

A general explanatory principle emerges from our conception of the externalization of language. If the externalization of language contains only language-specific phonology (Substance-free Logical Phonology) and non-language-specific phonetics (Cognitive Phonetics and other performance systems), with no language-specific phonetics, then we have a model that can clearly and consistently differentiate between **phonological** and **phonetic** aspects of sound patterns and processes: anything sound-related that is represented in the mind and is language-specific is phonological by definition; other aspects of speech, which are not language-specific and do not vary systematically between I-languages, are phonetic. Therefore, given a spoken utterance, we must ask: can the properties of that utterance be accounted for by transduction at the PPI or by the biomechanics of the vocal apparatus? If they can, then the properties are phonetic. If they can’t, then the properties are somehow encoded in the mind of the speaker and are thus phonological. For example, the partial nasalization of a vowel before a nasal consonant is due to the biomechanics of the velum and is thus a phonetic effect, i.e., a kind of anticipatory

[12] Note that the line of reasoning utilized by Bradlow (1995) is problematic only if we are investigating the architecture of the human language faculty as it is instantiated by each unique I-language. A vast amount of phonetic research *justifiably* conflates data from multiple I-language simply because it is not investigating the nature of I-language. Thus, our critique of Bradlow (1995) is a critique *only as it pertains to the nature of I-language*, and not a critique of the study in general.

coarticulation. We expect to find it in the spoken production of every I-language (Chafcouloff & Marchal 1999; see Cho & Ladefoged 1999: 213–214 for a more intricate example related to VOT).

It is worth pointing out that the use of the IPA adds a layer of complication to the kinds of analyses that we've been dealing with. For example, we transcribe as [i] anything that we judge as 'the high front unrounded vowel', but when we compare two I-languages that have a 'high front unrounded vowel', we often discover that these two [i]'s, while somewhat similar, still systematically differ in their realizations. Upon this observation, many scholars invoke LSP, as we see with Bradlow (1995) and many others: the *same* [i] in two different languages receives a language-specific phonetic interpretation, leading to two different acoustic spaces. We, in contrast, claim that these are in fact *not the same* vowels: they are *phonologically* different, despite the fact that we can phonetically describe both of them as high front unrounded and that we use the same symbol for their transcription. In this sense the use of the IPA to represent phonological reality is clearly inadequate.

Another complication is the fact that current versions of feature theory are not sufficiently developed so as to offer particular features that would account for all of these subtle-but-systematic differences between I-languages. That's what's still missing from our explanation of Bradlow's (1995) English vs. Spanish vowels: we have claimed that English [i] is featurally different from Spanish [i], but we didn't propose the exact feature (or features) that would differentiate between them; no current version of feature theory offers such a feature.[13] We interpret this deficiency as a strong impetus for further work on improving the feature theory, perhaps along the lines outlined in Hale et al. (2007), rather than trying to salvage a theoretically and empirically inadequate LSP. The positive prospect of this endeavor is reflected in the fact that 30 binary features (only slightly more than is usually assumed) with surface underspecification can yield over 206 trillion distinct surface segments, which is surely more than enough to capture all of the examples of language-specific *phonology* that have been (mis)interpreted as language-specific *phonetics*.

Note that we do not claim that *all* purported examples of language-specific phonetics can be explained with features. Instead, we think that the totality of alleged LSP phenomena should be re-analyzed as *either* (a) language-specific phonological phenomena (for which we offer the Logical Phonology model); *or* (b) non-phonological and therefore non-language-specific transduction of surface representations (for which we offer the Cognitive Phonetics model); *or* (c) systematic biomechanics of speech production, speech pragmatics, and random situational factors which modulate the acoustic output from the body. Phonetic variation and cross-linguistic differences are best explained by careful analysis along these three mutually exclusive dimensions.

[13] Perhaps the only difference between them is that the English [i] is [+ATR] (or [+TENSE], if one believes in the existence of that feature) while the Spanish [i] is underspecified for that feature.

5. Conclusion

Adopting the competence–performance dichotomy, strict internalism and modularity, we have argued that phonology (phonological competence) and phonetics (the sensorimotor system in charge of speech production and perception) are distinct and non-overlapping domains in the complex process of the externalization of language through speech. These two domains are connected via the phonology-phonetics interface (PPI): a two-way transduction system that converts between feature-based surface phonological representations and the temporally coordinated motor programs that we call phonetic representations.

Figure 4 summarizes the general architecture of the PPI in the theory of Cognitive Phonetics (CP). To connect the substance-free phonology with the substance-laden physiological phonetics, CP takes features of phonological surface representations (SRs) and relates them to neuromuscular activity via the paradigmatic transduction algorithm (PTA), and subsequently arranges that activity temporally via the syntagmatic transduction algorithm (STA), thus generating phonetic representations (PRs) which are directly interpretable by the sensorimotor (SM) system.

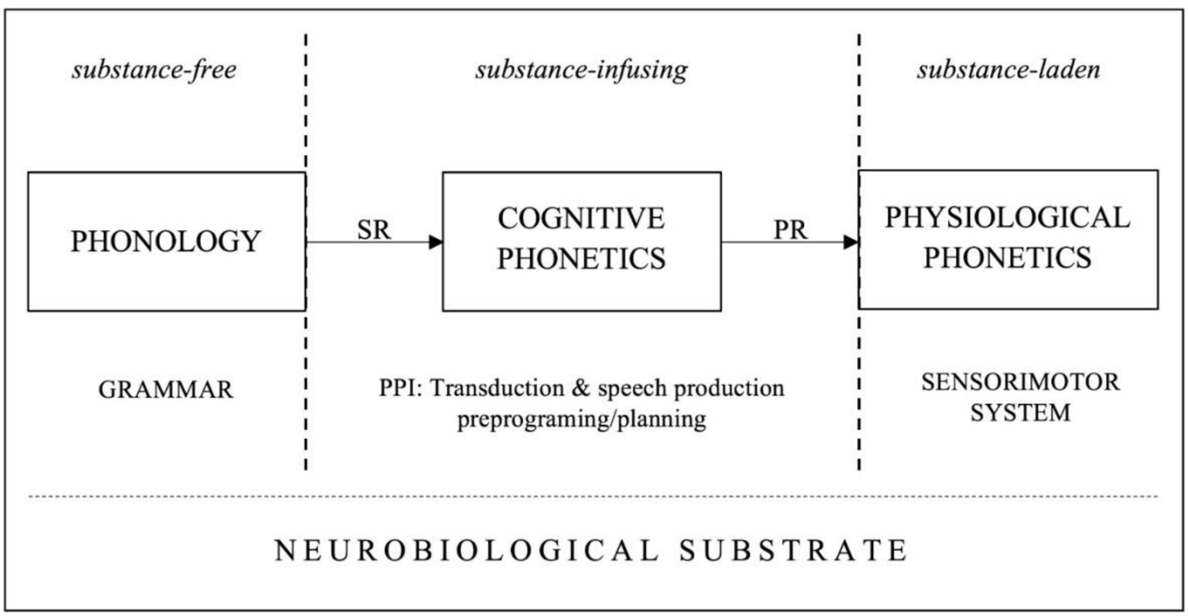

Figure 4. The architecture of the phonology-phonetics interface in the theory of Cognitive Phonetics.

Our conception of the externalization of language through speech therefore only has three steps: phonology, cognitive phonetics, and physiological phonetics. There is no language specific-phonetics (LSP). A research agenda for the future is to test if the various LSP patterns that have been proposed in the literature can be re-analyzed so as to conform to this framework.

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